

Threat Discernment Under Uncertainty

From control flow to biological evolution, environmental sensation, awareness, state integrity, and hazard recognition

Independent Perplexity AI Version

Concept and laboratory framework • not an operating protocol

Document field	Entry
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Classification	Conceptual research design; non-operational
Supersedes	Initial conversation notes and unsynchronized AI drafts; preserves them as provenance, not as approved doctrine

Purpose of this edition

This edition creates a stable, versioned foundation for deciding what the proposed laboratory is, what it is not, which claims are testable, and which decisions must be made before any implementation. It independently evaluates the supplied outputs from ChatGPT, Perplexity AI, Microsoft Copilot, Google Gemini, and Grok, but it does not adopt any statement merely because several systems repeated it.

Safety boundary

This document does not authorize procurement, preparation, dosing, release, exposure, handling, or disposal of poisons, toxins, corrosives, pathogens, or other hazardous agents. Early validation should use simulation, benign surrogates, prerecorded or synthetic cues, nonliving sensor platforms, and validated non-animal methods. Any later work involving hazardous chemicals, people, animals, ecosystems, or living tissues requires qualified institutional review, applicable legal approvals, containment, emergency planning, and domain-specific protocols.

Document Control

Version record

Version	Date	Preparer	State	Change record
0.0	Prior to 21 Sep 2026	Research Architect + multiple AI systems	Uncontrolled source material	Initial idea, follow-up constraints, and separate AI restatements.
0.1	21 Sep 2026	Perplexity AI	Foundational draft	Independent analysis; definitions; hypotheses; staged design; alternative analyses; decision gates; traceability.
0.2	Future	To be assigned	Planned	Incorporate owner decisions, expert review, entity class, endpoints, ethics route, and pilot design.
1.0	Future	To be assigned	Controlled baseline	Ratified charter and study architecture; still separate from operating SOPs.

Change discipline

- Every revision should retain the document identifier, increment the version, record the date, name the preparer and approver, and describe substantive changes.
- Source statements must remain distinguishable from interpretations, decisions, hypotheses, and empirical findings.
- A claim that a note was “saved” is not a persistence record. A version is controlled only when its file, checksum or repository revision, author, date, and status can be verified.
- Terminology changes must update the glossary, hypotheses, measures, and traceability matrix together.
- No conceptual charter should be treated as a chemical-hygiene plan, animal protocol, biosafety submission, engineering specification, or standard operating procedure.

Decision rights

The Research Architect owns the originating questions and decides the project’s intended scope. Scientific, ethical, safety, and regulatory authorities retain their own approval powers; conceptual ownership does not substitute for those approvals. Perplexity AI is the preparer of this independent draft, not a principal investigator, institutional sponsor, safety officer, veterinarian, toxicologist, or approving authority.

Reading convention

The terms “entity,” “environment,” and “threat” are intentionally broad at the conceptual level. Whenever an experiment is proposed, each must be replaced by a concrete operational definition. Bracketed numbers refer to the evidence notes near the end of the document.

Executive Determination

The project has a strong and coherent core: study how a bounded entity constructs action-relevant state estimates from incomplete, noisy, delayed, conflicting, and possibly deceptive observations, then distinguishes opportunity, neutrality, uncertainty, and potential harm. The most important improvement is to define the laboratory as a threat-discernment and state-reconciliation program—not initially as a “poison laboratory.” This keeps the scientific question primary and prevents one hazard class from silently determining the ontology, ethics, and equipment of the entire program.

The supplied AI drafts correctly converge on five principles: perception is not direct access to reality; sensory architecture varies among entities; threat is relational and context-dependent; non-detection does not establish safety; and a useful map can be probabilistic rather than perfectly true. Research on animal sensory diversity confirms that human-centered stimulus design can create confounds and false conclusions, while multisensory research shows that discrepancy resolution includes deciding whether cues share a cause before integrating them.[1][2]

The drafts are not yet a laboratory design. They lack a controlled vocabulary, entity and endpoint selection, explicit null hypotheses, a distinction between hazard and threat, a ground-truth protocol, an ethics route, preregistered analysis rules, and a plan for unknown or unobservable hazards. Those omissions must be resolved before experimental materials, facilities, or organisms are selected.

Recommended program sequence: formal model and simulation; benign-proxy testbed; nonliving sensor or robotic platform; validated in vitro or computational toxicology where relevant; and only then, if scientifically necessary and independently approved, organism-level studies. OECD integrated testing guidance supports combining mechanistic knowledge with alternative methods, while the 3Rs require replacement, reduction, and refinement where animal research is considered.[6][7]

Origin and Provenance

How the topic arose

The originating thought linked control flow with biological evolution and advancement, environmental sensation, awareness, and threat detection. It asked how an organism knows what surrounds it; how temporal integrity and presentation-state integrity affect that knowledge; how discrepancies among presented information, environmental conditions, and internal state are reconciled; how food, toxins, poisons, corrosives, and other dangers are distinguished; how a reliable map can be built from unreliable inputs; and how these capacities can be tested.

Two refinements changed the project materially. First, the laboratory must not assume a universal five-sense apparatus or treat human sensory categories as the default. Second, the term “threat” must not imply that every entity or environment is dangerous, that threats are constant, or that all threats are detectable, knowable, classifiable, avoidable, or included within the laboratory’s test domain.

Source set reviewed

Source	Distinct contribution	Treatment in this draft
Originator statements	Research purpose, anti-anthropocentric constraint, open-world threat limits.	Controlling source for intent; wording clarified but not overridden.
ChatGPT responses	Reliable-world-model framing; staged processing chain; open-world problem.	Retained as useful architecture; unsupported “saved” status not relied upon.
Perplexity AI responses	Strongest definitions, modular questions, integrity concepts, test directions, and laboratory limits.	Retained and tightened; distinctions and safety gates added.
Microsoft Copilot	Affordances, predictive/Bayesian options, relational threat framing, concise lab posture.	Used as an alternative lens; examples treated as illustrations, not evidence.
Google Gemini	Triangulation among presentation, environment, and internal baseline; explicit sensory constraints.	Retained after replacing “objective environment” with a reference-state model.
Grok	Compact record of constraints.	Useful provenance summary; insufficient as design basis.

Independent posture

This is not a consensus average. Agreement among language models can reflect shared training patterns rather than independent empirical confirmation. The framework therefore separates attractive formulations from testable constructs and uses external scientific and regulatory sources where factual support is needed.

Assessment of Drafts

What works

Element	Why it works	Design consequence
State estimation under uncertainty	Unifies sensing, memory, mapping, and action without requiring perfect perception.	Represent beliefs and confidence, not only categorical answers.
Relational threat	Prevents labeling an object as universally dangerous.	Index every threat claim to entity, state, context, time, exposure path, and endpoint.
Sensory pluralism	Avoids treating human vision/hearing/taste/smell/touch as universal.	Create an entity-specific sensorium profile before task design.[1]
Discrepancy reconciliation	Makes conflicting cues and model revision central rather than incidental.	Manipulate cue reliability, causal relatedness, delay, and contradiction.[2]
Temporal integrity	Highlights freshness, ordering, simultaneity, persistence, and causal attribution.	Timestamp observations and vary lag, jitter, duration, and event order.[3]
Open-world limits	Recognizes undetected, unknown, latent, and unaddressable hazards.	Include abstention/unknown responses and novelty trials; never equate “not detected” with “safe.”[10]
Action loop	Links perception to consequences and learning.	Measure both classification and policy quality across feedback cycles.

What does not yet work

Problem	Why it matters	Required correction
"Threat can be anything" is unlimited	Without a protected endpoint, almost any change can be called harm.	Define viability or value variables, unacceptable transitions, and time horizon.
Threat, hazard, stressor, exposure, and risk are conflated	A harmful capacity is different from contact, probability, and realized damage.	Use the layered ontology in this document.
"Environment states" implies direct ground truth	Experiments observe reference instruments and constructed conditions, not reality without error.	Define a reference environmental state with calibration and uncertainty.
"Awareness" is underspecified	It can mean responsiveness, access, reportability, self-modeling, or consciousness.	Use operational tiers and avoid consciousness claims unless separately justified.
Evolution and within-life learning are mixed	Evolutionary adaptation, development, learning, and moment-to-moment control operate on different timescales.	Model them as separate update loops.
Chemical classes dominate the framing	Poison, toxin, and corrosive are different categories and may not be sensed directly.	Separate environmental hazard characterization from entity discernment.[12]
No gold-standard design	Without independent state measurement, accuracy cannot be estimated.	Specify reference sensors, calibration, blind coding, and uncertainty bounds.
No ethics or safety architecture	Real exposure could cause avoidable harm and confound behavior with injury.	Begin with safe proxies and alternatives; require independent approvals before escalation.[6][7]
No inferential plan	A response can arise from unintended cues, side channels, stress, or experimenter effects.	Preregister contrasts, randomize, blind where possible, and run cue-isolation controls.[1][8][9]
"Saved" was asserted inconsistently	Conversation text is not version control.	Use controlled artifacts, revisions, ownership, status, and checksums.

What must be addressed first

1. Choose the first entity class: simulated agent, robot/sensor platform, cell/tissue model, microorganism, plant, invertebrate, vertebrate, human participant, or collective system.
2. Choose the protected variables and adverse endpoints: survival, tissue integrity, homeostasis, performance, reproduction, resource access, mission completion, or another explicit criterion.
3. Decide whether the initial program studies detection, discrimination, inference, learning, mapping, action selection, or a staged combination. These are not interchangeable outcomes.
4. Define "presentation" and the presentation channel. A cue presented by the apparatus may differ from the stressor, exposure, or environmental state it purports to indicate.
5. Specify the initial hazard representation. The default should be simulation or harmless proxy, not active toxic or corrosive exposure.
6. Define the reference-state instrumentation, calibration, uncertainty budget, synchronized clock, and provenance record.
7. Select the alternative theories to compare and define observations that would discriminate among them.
8. Establish governance: study owner, scientific lead, ethics reviewer, chemical safety lead, data steward, stopping authority, and version approver.

Purpose and Scope

Main purpose

The program's purpose is to determine how entities with different sensing and control architectures acquire, time-align, interpret, and act on uncertain environmental and internal information; how they distinguish known hazard, apparent hazard, opportunity, neutrality, and unknown conditions; and under what conditions their internal maps remain reliable enough for action.

Primary unit of analysis

The primary unit is not an isolated stimulus and not an object labeled "threat." It is an entity–environment relation over time: a bounded entity with particular vulnerabilities, capacities, internal conditions, sensors, memory, and actions interacting with a dynamic environment through one or more exposure and information pathways.

In scope

- Biological organisms, artificial agents, nonliving sensors, and hybrid systems, provided each study specifies which category is under examination.

- Exteroception, interoception, proprioception, chemical and physical sensing, distributed sensing, instrument-mediated sensing, and synthetic modalities.
- Known, ambiguous, latent, delayed, cumulative, deceptive, unobservable, and novel hazard conditions.
- Food/resource discrimination, hazard avoidance, state estimation, map formation, confidence, learning, and action selection.
- Simulation, behavioral tasks, robotics, validated in vitro systems, field observation, and ethically approved organism research.

Out of scope for v0.1

- A claim to enumerate all threats or certify an environment as universally safe.
- An assumption that every response implies conscious awareness or an explicit internal representation.
- Operational procedures for hazardous-agent preparation, dosing, containment, exposure, or disposal.
- Medical diagnosis, human-subject intervention, autonomous safety certification, or regulatory chemical classification.
- A universal model that erases species, developmental, individual, ecological, or technological differences.

Core Ontology

Terms

Term	Foundational definition
Entity	A bounded biological, artificial, collective, or hybrid system whose state, capacities, sensors, and outcomes can be specified for a study.
Environment	The external conditions and processes that can interact causally or informationally with the entity within a declared spatial and temporal boundary.
Reference environmental state	The best independently measured account of task-relevant environmental variables, including instrument error and unresolved uncertainty; not metaphysical “ground truth.”
Internal state	Task-relevant physiological, computational, developmental, motivational, or structural variables of the entity.
Presentation	The signal pattern made available at an interface to the entity. It may be generated by the environment, an apparatus, another entity, or the entity itself.
Observation	A presentation after transduction by a sensor and any preprocessing; it is already selective and potentially noisy.
Stressor	A physical, chemical, biological, informational, or relational factor capable of perturbing the entity or its environment.
Hazard	An intrinsic or context-conditioned capacity of a stressor or condition to cause a defined adverse effect.
Exposure	Contact or co-occurrence between entity and stressor through a specified path, magnitude, duration, frequency, and timing.[4][5]
Harm	A realized adverse change relative to a declared protected variable, viability range, or value criterion.
Threat	A context-indexed possibility that a condition or process could produce harm to a particular entity through a plausible exposure/action pathway within a relevant time horizon.
Risk	An estimate or characterization combining uncertainty about occurrence/exposure with the nature and magnitude of adverse consequences; it is not identical to threat.
Affordance	An action possibility arising from the relation between environmental features and an entity's capacities; the same feature may afford approach, use, avoidance, or nothing to different entities.[13]
Map	Any persistent or transient structure—spatial, chemical, temporal, causal, relational, or policy-based—that supports prediction and action. It need not be pictorial or consciously accessible.

Threat relation

A threat claim should have the form: condition C is a threat to entity E, in context K, during interval T, relative to protected variable V, through pathway P, with uncertainty U. Removing those indices turns a testable relation into an unrestricted label.

A practical representation is: Threat = plausible adverse transition; Risk = characterized likelihood/exposure and consequence of that transition. The laboratory should preserve “unknown” when either side cannot be estimated rather than forcing a safe/unsafe binary.

Threat-state matrix

Reference condition	Entity judgment	Interpretation	Laboratory implication
Hazardous and actionable	Threat	Correct detection with potential control	Measure action quality separately from detection.
Hazardous but unobservable	No threat / unknown	False negative or justified abstention, depending on available evidence	Do not attribute failure without verifying detectability.
Hazardous and observable	No threat	Miss	Estimate sensitivity and causal source of failure.
Nonhazardous	Threat	False alarm or rational precaution under asymmetric costs	Interpret relative to payoff and uncertainty.
Ambiguous reference state	Unknown	Calibrated uncertainty	Reward abstention when evidence is insufficient.
Hazard recognized but no feasible response	Threat	Detection without addressability	Do not treat lack of avoidance as lack of recognition.
Adaptive action without reportable judgment	Implicit/unknown	Competent control without demonstrated explicit representation	Separate behavior from awareness claims.

Integrity Constructs

Temporal integrity

Temporal integrity is the degree to which observations, internal states, environmental changes, actions, and outcomes remain correctly associated with their acquisition time, ordering, duration, persistence, and causal window. It includes clock synchronization, latency awareness, freshness, sequence, temporal binding, and the ability to revise beliefs when information becomes stale. Multisensory temporal binding is probabilistic: signals arriving within a window may be treated as one event even when asynchronous, and those windows can vary and adapt.[3]

Dimensions

- Timestamp integrity: when was the observation acquired, transformed, received, and used?
- Order integrity: which event preceded which action or outcome?
- Freshness integrity: does the observation still describe the current condition?
- Duration integrity: was exposure brief, repeated, chronic, or cumulative?
- Cross-channel alignment: do different modalities refer to the same interval?
- Attribution window: which prior event is treated as the cause of a delayed effect?
- Memory integrity: was the retained record altered, generalized, or misbound?

Presentation-state integrity

Presentation-state integrity is the degree to which the presentation available to an entity preserves task-relevant relations to the reference environmental state and the entity's current interaction context. It is not visual fidelity and not perfect correspondence. A presentation can be low-fidelity yet action-sufficient, or physically accurate yet unusable to the entity.

Dimensions

- Coverage: are task-relevant variables represented at all?
- Fidelity: how closely do represented values track the reference state?
- Provenance: what generated, transformed, filtered, delayed, or relayed the presentation?
- Causal validity: does the cue arise from the condition it is taken to indicate?
- Cross-modal coherence: do channels agree because of a shared cause, correlated noise, or deliberate coupling?
- Entity accessibility: can the entity transduce and discriminate the presentation?
- Tamper/deception resistance: can mimicry, camouflage, spoofing, occlusion, or apparatus artifacts produce misleading evidence?

State reconciliation

Reconciliation is not simple averaging. The entity must determine whether observations belong to the same cause, estimate each source's reliability, compare external observations with internal state and memory, preserve incompatible hypotheses when necessary, and select an action under asymmetric costs. Human multisensory studies support reliability-weighted integration while also showing that the system must infer whether cues should be fused or segregated.[2]

Minimal reconciliation cycle

1. Acquire observations and internal-state signals with time and provenance metadata.
2. Estimate quality, detectability, missingness, and likely corruption for each channel.
3. Infer causal grouping: which signals refer to the same source or event?
4. Generate competing state hypotheses rather than one forced interpretation.
5. Update confidence using observations, priors, memory, and context.
6. Classify action relevance: opportunity, neutral, possible hazard, known hazard, or unknown.
7. Choose act, inspect, sample, wait, avoid, seek help, or abstain according to consequences and capability.
8. Observe feedback, attribute it cautiously, and update the map, sensor trust, and policy.

Foundational Propositions

ID	Proposition	Falsifiable implication
P1	Perception is entity-relative and sensor-mediated.	Changing the sensorium or cue domain can change performance without changing the environmental condition.
P2	Useful state estimation need not be perfectly veridical.	An entity may act effectively with a compressed or biased map if task-relevant relations are preserved.
P3	Threat is relational, not an intrinsic universal label.	The same condition produces different threat judgments when entity state, vulnerability, dose, pathway, or available action changes.
P4	Non-detection is not evidence of absence unless detectability is established.	Performance near chance under an inaccessible cue does not demonstrate failed reasoning.
P5	Cue integration requires causal inference.	Increasing discrepancy can shift behavior from fusion to segregation rather than producing simple weighted averaging.
P6	Temporal mismatch can create state mismatch.	Controlled delays can induce false binding, stale-map actions, or incorrect causal learning.
P7	Internal state changes environmental meaning.	The same external condition can produce different choices across hunger, injury, developmental stage, or mission state.
P8	Uncertainty is an output, not merely a defect.	Well-calibrated abstention can outperform forced classification in novel conditions.
P9	Addressability is independent of detectability.	An entity may correctly detect a hazard yet lack an effective action.
P10	Open-world competence requires handling unknowns.	Closed-set accuracy can remain high while novel-hazard performance fails.[10]

Hypotheses

Primary hypothesis

H1 — Adaptive discernment hypothesis: An entity's ability to maintain viable action under uncertain conditions depends on the joint quality of sensor accessibility, temporal integrity, causal attribution, internal-state estimation, uncertainty calibration, and available actions—not on sensory count or raw detection accuracy alone.

Null H1■: After controlling for signal strength and task difficulty, temporal integrity, causal attribution, internal state, confidence calibration, and action availability do not improve prediction of viable action beyond raw detection accuracy.

Secondary hypotheses

ID	Hypothesis	Primary contrast	Candidate endpoint
H2	Reliability-weighted reconciliation: entities give more weight to channels that are more reliable, but only when cues are inferred to share a cause.	Congruent vs discrepant cues across graded reliability.	Cue weights; fusion/segregation probability; localization/classification error.
H3	Temporal-integrity degradation: lag, jitter, reordering, or staleness degrades attribution and map updating before it necessarily degrades simple detection.	Synchronized vs delayed/reordered channels.	Attribution error; stale-map error; recovery time.
H4	Presentation-integrity degradation: deceptive or transformed presentations reduce performance according to entity-specific accessibility, not human-perceived fidelity.	Species/entity-calibrated cue vs human-calibrated cue.	Discrimination, false alarm, missed detection, learning rate.
H5	Internal-state modulation: hazard/resource appraisal changes with physiological or computational state even when external presentation is held constant.	State A vs State B under identical external cues.	Choice, threshold, latency, confidence.
H6	Active sensing advantage: entities allowed to sample, move, or query resolve ambiguity more accurately than passive observers, conditional on sampling cost.	Active vs yoked passive observation.	Information gain, error, action cost, regret.
H7	Unknown-aware advantage: systems with an explicit unknown/abstain option are better calibrated on novel conditions than forced closed-set classifiers.	Known-class only vs unknown-enabled evaluation.	Open-set error, calibration, unsafe acceptance rate.
H8	Redundancy is conditional: additional modalities improve robustness only when they add independent information or expose corruption.	Single vs redundant correlated vs redundant independent channels.	Robustness under dropout/spoofing; mutual information; task success.
H9	Actionability dissociation: detection and adaptive outcome separate when the entity lacks an effective response.	Response available vs blocked.	Detection unchanged; outcome diverges.

Research Questions

Sensing and access

- Which variables can the entity transduce directly, indirectly, instrumentally, or not at all?
- What are each channel's range, resolution, thresholds, noise distribution, latency, adaptation, saturation, and failure modes?
- Which cues are salient to the entity but invisible or irrelevant to the experimenter?
- Does the apparatus accidentally provide side channels such as heat, vibration, odor, polarization, refresh flicker, electromagnetic fields, or operator behavior? Research on animal experimentation shows that such human-overlooked cues can reverse interpretations.[1]

Reconciliation and maps

- What counts as agreement or conflict among channels, memory, internal state, and reference measurements?
- How are sources grouped into common-cause versus separate-cause hypotheses?
- How are old observations discounted in a changing environment?
- Does the map encode metric space, topological routes, chemical gradients, event transitions, affordances, policies, or combinations?
- How does the entity represent confidence, ambiguity, novelty, and missingness?

Threat and resource appraisal

- Which protected variables define adverse change for this entity?
- Is the entity detecting the hazardous property, a correlated cue, a learned label, a bodily consequence, or another entity's response?
- How do dose, route, duration, cumulative exposure, life stage, and internal condition change classification?
- When are false alarms adaptive because misses carry greater cost?
- Can the entity distinguish “not hazardous,” “not currently exposed,” “not detectable,” “unknown,” and “recognized but unaddressable”?

Learning and evolution

- Which responses are innate, developmentally canalized, individually learned, socially learned, or externally programmed?
- Does feedback improve the causal model, only the action policy, or both?
- How do changes within an individual differ from population-level selection across generations?
- What trade-offs—energy, speed, false alarms, sensor cost, or missed opportunity—constrain the architecture?

Validation

- What independent evidence supports the reference environmental state?
- Can performance replicate across apparatuses, operators, contexts, and laboratories?
- What result would falsify each proposed mechanism rather than merely show task success?
- How will the study detect unintended cues, leakage, experimenter influence, and post hoc reinterpretation?

Laboratory Architecture

Recommended name

Use “Threat Discernment and State Integrity Laboratory” as the umbrella name. Treat “poison, toxin, and corrosive test domain” as one application module rather than the identity of the entire laboratory. This preserves the broader science and allows nonchemical threats, resource opportunities, neutral conditions, and no-threat baselines.

Two coupled systems

System	Purpose	Required separation
Environmental characterization system	Creates or measures the reference condition, timing, exposure path, and uncertainty.	Must not infer entity perception from apparatus settings alone.
Entity discernment system	Measures observation, judgment, behavior, internal-state change, learning, and action.	Must not infer actual hazard solely from the entity's response.

Coupling the two systems without conflating them is essential. A cue can be detected without exposure; exposure can occur without a detectable cue; harm can occur after delay; and avoidance can happen for reasons unrelated to the intended hazard.

Control-flow model

Reference environment → presentation channels → entity-specific transduction → time/provenance buffer → cue-quality estimation → causal grouping → external/internal state estimation → discrepancy register → hazard/resource/unknown appraisal → action selection → environmental change → outcome and learning.

Parallel loops operate on different timescales: fast regulation, action-perception cycles, within-life learning, development, social or collective adaptation, and population evolution. Results from one loop must not be reported as evidence for another without an explicit bridge.

Entity sensorium profile

Domain	Questions to record
Physical access	What can reach the entity: photons, molecules, vibration, pressure, temperature, electric or magnetic fields, radiation, contact, data packets?
Receptors/transducers	Which structures or devices convert conditions into signals? Are receptors known, inferred, distributed, or external?
Operating range	Thresholds, dynamic range, resolution, saturation, adaptation, recovery, noise, and damage limits.
Temporal properties	Sampling rate, latency, integration window, refractory period, circadian/seasonal phase, and clock stability.
Spatial properties	Field of view, directional sensitivity, gradient sensitivity, range, occlusion, and localization uncertainty.
Preprocessing	Filtering, normalization, compression, feature extraction, sensor fusion, and missing-data handling.
Internal sensing	Homeostatic, proprioceptive, energetic, structural, or diagnostic signals available to the entity.
Action repertoire	Approach, avoid, inspect, sample, communicate, wait, shield, repair, reject, or abstain.
Ecological relevance	Whether stimuli, timing, setting, and response are within the entity's natural or designed operating regime.[1]

Staged Research Program

Stage 0 — Governance and formalization

- Ratify definitions, entity boundary, protected variables, endpoints, exclusions, and stopping rules.
- Create a hazard and ethics review pathway separate from scientific enthusiasm.
- Specify data provenance, clock synchronization, calibration, blinding, randomization, preregistration, and change control.
- Set the escalation rule: later stages proceed only if the question cannot be answered adequately at a lower-risk stage.

Stage 1 — Simulation

- Use partially observable environments with controlled noise, delay, missingness, contradiction, novelty, and changing causal structure.
- Compare mechanisms: fixed thresholds, Bayesian causal inference, state-space filters, active inference, reinforcement learning, affordance-based control, and unknown-aware classifiers.
- Retain exact latent-state histories so map error, calibration, and causal attribution can be scored without measurement ambiguity.

Stage 2 — Benign physical testbed

- Use nonhazardous surrogates whose cues can be independently manipulated: color, odor substitute, conductivity, pH-indicating signals without exposure, temperature within safe limits, texture, vibration, or machine-readable synthetic channels.
- Use robotic or nonliving sensor entities first to validate timing, cue isolation, state logging, and open-world trials.
- Deliberately decouple cue from condition to distinguish direct detection from learned association.

Stage 3 — Non-animal biological alternatives

- Where chemical adversity is relevant, prefer validated in vitro, organoid, tissue, cellular, computational, read-across, or AOP/IATA approaches appropriate to the endpoint. OECD guidance frames AOPs and alternative methods as components of integrated assessment rather than one universal assay.[6]
- Keep toxicological effect characterization distinct from organism-level awareness or behavioral discernment claims.

Stage 4 — Organism studies, conditional

- Proceed only when the scientific question requires an intact organism and cannot be answered by lower-risk methods.
- Apply Replacement, Reduction, and Refinement; obtain all relevant institutional and legal approvals; use predefined humane endpoints and independent stopping authority.[7]
- Follow species-specific sensory ecology, housing, circadian, social, and behavioral requirements; a human-convenient apparatus may be inaccessible or stressful to the test species.[1]
- Randomize allocation and order, blind outcome assessment where feasible, disclose exclusions, and report enough detail for replication. ARRIVE identifies these practices as central to transparent in vivo reporting.[8][9]

Experiment Families

Calibration and detectability

Goal: establish whether the entity can access the cue under the exact apparatus conditions before interpreting failure as poor judgment. Manipulate one channel at a time across safe, preapproved ranges; estimate threshold, psychometric or response curve, latency, saturation, adaptation, and individual variation. Include sham, blank, positive-control, and side-channel controls.

Cue isolation and substitution

Goal: determine what the entity is actually using. Hold the reference condition constant while removing, masking, or substituting candidate cues; then hold the cue constant while changing the condition. A complete design includes congruent, incongruent, cue-only, condition-only, and neither conditions.

Cross-channel conflict

Goal: test integration, segregation, and causal inference. Vary cue reliability and discrepancy independently. Compare forced-fusion models, winner-take-all rules, reliability weighting, model averaging, and explicit “separate causes” responses.[2]

Temporal integrity

Goal: test stale information, order, binding, and delayed outcomes. Manipulate channel lag, jitter, reordering, missing intervals, and environmental change rate while preserving overall signal quantity. Distinguish detection error from incorrect time assignment and incorrect cause assignment.[3]

Internal-state dependence

Goal: test whether identical external information is interpreted differently across controlled internal states. Use only ethically and scientifically justified state manipulations; measure the internal state rather than assuming it from schedule or treatment. Analyze state as a moderator, not a nuisance to be averaged away.

Map construction and remapping

Goal: determine what kind of map supports action. Change landmarks, gradients, transitions, resource locations, or cue reliability; compare route performance, generalization, uncertainty, and remapping speed. Include probe trials that separate memorized actions from inferred structure.

Unknown and novelty

Goal: test whether the entity can refrain from unsafe generalization. Introduce novel but benign classes, unseen cue combinations, sensor failures, and out-of-distribution conditions. Open-world research warns that “unknown” is not equivalent to low confidence: a system may be confidently wrong on an unseen class.[10]

Actionability

Goal: separate recognition from control. Provide, withhold, or vary response options after the same detectable condition. Measure whether the entity signals, waits, samples, avoids, or persists; do not infer nonrecognition from failure to escape an inescapable condition.

Learning and transfer

Goal: identify what changes after feedback. Test acquisition, retention, extinction, reversal, context transfer, cross-modal transfer, and recovery after sensor corruption. Separate a learned association with a cue from causal knowledge of the underlying hazard.

Experimental Factor Matrix

Factor	Levels/examples	Reason
Entity class	Simulated, robotic, cellular, plant, invertebrate, vertebrate, human, collective.	Prevents generalizing across incompatible architectures.
Sensor availability	Native, augmented, degraded, absent, unknown.	Tests access before inference.
Reference condition	Opportunity, neutral, possible hazard, hazardous, indeterminate.	Avoids threat-only environments.
Cue relation	Direct, correlated, learned, deceptive, absent, contradictory.	Distinguishes condition from presentation.
Observability	Fully, partially, indirectly, or unobservable.	Creates principled limits on claims.
Timing	Synchronous, delayed, jittered, reordered, stale, cumulative.	Tests temporal integrity.
Environment dynamics	Stable, slowly changing, volatile, adversarial.	Tests model revision.
Internal state	Baseline and prespecified alternative states.	Tests relational threat and motivation.
Novelty	Known instance, new instance, new class, new modality, sensor fault.	Tests open-world behavior.
Actionability	Effective action, costly action, ambiguous action, no available action.	Separates detection from outcome.
Feedback	Immediate, delayed, partial, deceptive, absent.	Tests attribution and learning.

Measures and Analysis

Primary measures

Construct	Measures	Interpretive caution
Detectability	Hit rate, false-alarm rate, sensitivity index, threshold, response latency.	A low response rate may reflect motor or motivational limits, not sensory failure.
Discrimination	Confusion matrix, balanced accuracy, class-conditional error.	Closed-set accuracy says little about novel conditions.
Calibration	Confidence–accuracy relation, Brier score, calibration slope, abstention quality.	Confidence must be behaviorally or computationally defined for the entity.
Temporal integrity	Order error, binding curve, state-state error, delay tolerance, recovery time.	Clock and apparatus latency must be measured independently.
Presentation integrity	Cue–state fidelity, side-channel dependence, spoof acceptance, corruption detection.	The entity may use cues the researcher did not intend.
Map quality	Localization error, route efficiency, prediction error, topological consistency, remapping speed.	Choose measures that match the map type; do not assume a Euclidean map.
Decision quality	Expected loss, regret, success under asymmetric costs, safe failure rate.	Optimal policy depends on the declared loss function, not accuracy alone.
Learning	Trials to criterion, retention, reversal cost, transfer, extinction.	Improvement may reflect cue memorization rather than causal understanding.
Robustness	Performance under dropout, noise, spoofing, context shift, and novel classes.	Stress tests must remain within ethical and equipment limits.

Analysis commitments

- Define the experimental unit and avoid pseudoreplication; repeated observations do not automatically create independent samples.
- Predefine exclusions, primary endpoints, contrasts, model-selection criteria, stopping rules, and missing-data treatment.
- Randomize treatment, trial order, apparatus position, and operator assignment where appropriate; blocking may be needed for day, cage, device, operator, or location effects.[8]
- Blind assessors and analysts where possible, and document who knew allocation at each stage.[9]
- Estimate effect sizes and uncertainty intervals, not only thresholded significance.
- Use held-out contexts and genuinely novel classes to evaluate generalization and open-world behavior.
- Compare mechanistic models on out-of-sample prediction, calibration, complexity, and intervention tests—not narrative fit alone.
- Report negative and indeterminate findings. “No observed effect under these tested conditions” is not “no capacity” and not “no threat.”

Core scorecard

Question	Minimum evidence
Could the entity sense the relevant cue?	Independent detectability calibration in the same apparatus and state range.
Did it identify the condition?	Performance above a preregistered baseline with alternative cues controlled.
Did it identify a threat?	Evidence that judgment tracks a defined adverse relation, not merely salience or aversion.
Did it reconcile discrepancies?	Behavior or model state changes systematically with cue reliability, causal structure, and internal state.
Did it build a map?	Transfer, prediction, route flexibility, or remapping beyond memorized stimulus-response chains.
Was uncertainty appropriate?	Calibrated confidence or abstention, especially on ambiguous and novel trials.
Was the response adaptive?	Outcome evaluated under a declared cost/loss structure and available actions.

Alternative Analyses

Why plural analysis is necessary

The same successful behavior can support different explanations. An entity may avoid a location because it inferred a toxic condition, memorized a cue, followed a gradient, copied another entity, reacted reflexively, or exploited an apparatus artifact. The program should therefore maintain competing models and design interventions that separate them.

Lens	Central claim	Best test	Failure risk
Bayesian causal inference	The entity estimates latent causes and reliability, integrating or segregating cues probabilistically.	Cross reliability with discrepancy; compare fusion, segregation, and model averaging.[2]	Good descriptive fit may not prove the nervous system implements Bayesian computation.
Signal detection / decision theory	Responses depend on discriminability and a criterion shaped by costs, priors, and payoffs.	Manipulate prevalence and miss/false-alarm costs while holding signal distributions stable.	Can describe choices without explaining learning or representation.
Predictive processing / active inference	Perception and action reduce mismatch between predicted and sensed states while maintaining viable internal conditions.	Manipulate controllability, expected precision, and active sampling.	Broad formulations can become difficult to falsify unless models are explicit.[11]
Ecological / affordance-based	Entities perceive action possibilities relative to body and environment rather than reconstructing a detached world model.	Change action capacity while preserving environmental geometry or cue.	May under-specify internal inference when delayed or hidden causes matter.[13]
Reinforcement learning	Values and policies are updated through experienced consequences.	Compare model-free, model-based, and latent-state learning under reversal and transfer.	Reward history can fit behavior without establishing threat concepts.
Dynamical systems / control	Adaptive behavior emerges from coupled entity-environment dynamics and feedback stability.	Perturb feedback gain, delay, observability, and controllability.	State variables may be hard to identify uniquely.
Ethology / sensory ecology	Performance depends on species-specific sensing, ecology, lifecycle, and natural behavior.	Replicate across ecologically valid and human-convenient apparatuses.[1]	Naturalism alone does not identify mechanism.
Open-world recognition	The system must identify unknowns rather than force every input into known classes.	Use novel classes, adversarially similar benign cases, and sensor faults.[10]	Unknown detection remains technically unsolved in general.
Adverse outcome pathway	Chemical exposure can be organized from molecular initiation through key events to adverse outcome.	Use mechanistically linked assays and weight-of-evidence integration.[6][12]	AOPs characterize adversity, not necessarily an organism's awareness or discernment.

Triangulation rule

No single lens should be declared the laboratory's universal theory. Each study should identify at least two plausible explanations, specify discriminating predictions in advance, and report when data underdetermine the mechanism. EPA risk-characterization guidance similarly calls for explicit uncertainty, reasonable alternative interpretations, and separation of scientific conclusions from policy judgments.[5]

Chemical Domain Boundaries

Separate the terms

"Poison," "toxin," and "corrosive" should not be used as interchangeable labels. A toxin is commonly reserved for a poisonous substance produced by a living organism; poison is a broader harm-causing substance under relevant exposure; and corrosivity refers to destructive chemical action on living tissue or materials, depending on the classification context. OSHA distinguishes health-hazard classes and defines skin corrosion as irreversible tissue damage, while corrosion to metals is a separate physical-hazard category.[12]

Detection is not toxicity

An entity may detect an odor, color, pH-associated cue, irritation, tissue damage, another organism's alarm response, or a learned symbol rather than the harmful agent itself. Conversely, a harmful chemical may produce no accessible cue before injury. Every study must state whether it evaluates environmental detection, predictive cue learning, exposure, early biological perturbation, adverse outcome, or post-harm response.

Safe initial strategy

- Represent threat relations with simulation and benign surrogates before considering active hazards.
- Validate the apparatus with nonliving sensors and independent reference measurements.
- Use validated alternative toxicology methods and integrated evidence where the question is chemical adversity.[6]
- Do not deliberately injure an entity merely to prove that it can detect injury; use the least harmful scientifically adequate method.
- Do not infer universal safety, toxicity, or detectability from one entity, modality, dose, route, life stage, or timescale.

Ethics, Safety, and Governance

Non-negotiable principles

- Proportionality: scientific value and necessity must justify residual risk after alternatives and controls.
- Separation of duties: the person advocating an experiment should not be the sole safety or ethics approver.
- Least-harm escalation: simulation and non-animal alternatives precede organism exposure unless a competent review body documents why they cannot answer the question.
- Humane and fail-safe design: stopping rules, rescue actions, exposure prevention, and emergency response are specified before testing.
- Uncertainty disclosure: unknown hazards, sensor blind spots, and limits of detection remain visible in reports and interfaces.
- No autonomous release: a classifier's "safe" label never authorizes uncontrolled human, animal, or environmental exposure.

Required roles before implementation

Role	Responsibility
Scientific lead	Own hypotheses, methods, analysis, and interpretation.
Entity-domain expert	Validate sensory, behavioral, physiological, or technical assumptions for the entity.
Toxicology / hazard expert	Define hazard classes, endpoints, exposure logic, and alternatives.
Chemical hygiene / biosafety lead	Approve facilities, containment, handling, emergency response, and waste processes.
Ethics authority	Review human, animal, ecological, or sentience-related concerns as applicable.
Statistician / methods lead	Define design, power, randomization, blinding, and model comparison.
Instrumentation lead	Maintain calibration, synchronization, reference measurements, and uncertainty budgets.
Data steward	Maintain provenance, access controls, metadata, retention, and version history.
Independent stopping authority	Pause or terminate work when predefined safety or welfare limits are reached.

Decision Gates

Gate	Question	Evidence required	Outcome
G0 — Concept	Is the question coherent and bounded?	Approved ontology, purpose, entity class, protected variables, and exclusions.	Revise or proceed to modeling.
G1 — Model	Can simulation distinguish mechanisms?	Preregistered models, identifiability analysis, synthetic only where explicitly labeled, and falsification criteria.	Refine or proceed to benign testbed.
G2 — Apparatus	Does the platform present and measure intended variables?	Calibration, timing audit, side-channel survey, reference-state uncertainty.	Repair or approve pilot.
G3 — Pilot	Are measures reliable and procedures safe?	Positive/negative controls, repeatability, stopping criteria, no unexpected harm.	Stop, redesign, or expand.
G4 — Biological need	Is intact-organism work necessary?	Alternatives assessment, scientific justification, 3Rs plan, ethics and safety approvals.[7]	Reject, defer, or authorize limited protocol development.
G5 — Interpretation	Do results support the claimed construct?	Competing-model comparison, robustness, uncertainty, alternative explanations.	Narrow claim, replicate, or advance.
G6 — Generalization	Does the result transfer?	Independent context, apparatus, entity, or laboratory replication.	Keep local or expand scope.

Immediate Questions

Questions for the owner

1. What is the first concrete entity class, and why is it the least risky informative starting point?
2. Which current-state variables define integrity or viability for that entity?
3. Is the first target a theory of perception, a test platform, an AI/robotics architecture, a comparative biology program, or a chemical-safety application?
4. What does "awareness" need to mean for this project: reactivity, integrated access, reportability, self-monitoring, or consciousness?

5. What counts as a map, and what behavior would distinguish a map from a memorized policy?
6. Which discrepancies matter first: cross-sensory conflict, sensor versus environment, present versus memory, internal versus external state, or prediction versus outcome?
7. What harm endpoint and time horizon define the first threat relation?
8. Should the system be optimized for low misses, low false alarms, calibrated abstention, or minimum expected loss?
9. What kinds of unknowns must be preserved rather than classified?
10. What evidence would cause the program to conclude that the proposed construct is unnecessary or incorrectly framed?

Questions for domain experts

- Which candidate cues are actually accessible to the selected entity, and which apparatus artifacts are likely?
- What lower-risk proxy preserves the information problem without reproducing the physical harm?
- Which endpoint can be measured before irreversible injury?
- Which individual, developmental, circadian, social, or ecological variables must be blocked or stratified?
- Which alternative methods are validated for the chemical or biological endpoint?
- Which approvals, facilities, personnel competencies, and emergency plans are mandatory?

Recommended First Study

Benign discrepancy-and-mapping pilot

Begin with a simulated or robotic entity in a controlled arena containing resource, neutral, and “hazard-proxy” zones. Give the entity two or more synthetic channels with independently controllable reliability and latency. No actual toxic or corrosive agents are needed. The entity may inspect, move, wait, abstain, or choose a route; the environment logs complete latent state and all channel transformations.

Manipulations

- Cue reliability: high, medium, low, missing.
- Cue relationship: congruent, discrepant, deceptive, and unrelated causes.
- Timing: synchronized, delayed, jittered, reordered, and stale.
- Internal state: different resource needs or sensor-health states encoded transparently.
- Novelty: unseen cue combinations and new benign classes.
- Actionability: direct route, costly detour, inspect option, or no effective response.

Predictions

- A robust entity should not always fuse discrepant channels; it should condition integration on inferred common cause and reliability.
- A temporally aware entity should discount stale information and recover after synchronization is restored.
- An unknown-aware entity should abstain or gather information on novel conditions rather than force a familiar label.
- A relational threat model should change policy when internal needs or action capacities change, even under the same external presentation.

Advancement criterion

Advance only after the task demonstrates repeatable measurement, identifiable mechanisms, calibrated unknown handling, and clear separation among cue, condition, exposure proxy, internal state, and outcome. A biological pilot should not be used to debug basic timing, logging, or classification errors.

Acceptance Criteria

Version 0.1 is successful if reviewers can answer the following without relying on oral explanation:

- What is being studied, and what is explicitly not being claimed?
- How are entity, environment, presentation, reference state, hazard, exposure, threat, risk, harm, map, and integrity distinguished?

- Which hypotheses are falsifiable, and what are their nulls or alternatives?
- Why does the first experiment not require real poison, toxin, or corrosive exposure?
- How will detectability, discrepancy reconciliation, uncertainty, mapping, and action quality be measured separately?
- Which alternative theories could explain the same behavior?
- Which decisions and approvals remain unresolved?
- How will subsequent revisions preserve provenance and avoid overstating evidence?

Traceability Matrix

Originating concern	Design response	Primary section
How does an organism know what is around it?	Entity sensorium profile, detectability calibration, state-estimation loop.	Core Ontology; Laboratory Architecture
Temporal integrity	Timestamp, lag, binding, staleness, causal-attribution constructs and trials.	Integrity Constructs; Experiment Families
Presentation-state integrity	Reference-state fidelity, provenance, causal validity, accessibility, and spoof tests.	Integrity Constructs
Reconcile presentation, environment, and current state	Competing hypotheses, reliability, causal grouping, internal-state moderation.	State Reconciliation; Alternative Analyses
Recognize threat, food, toxin, poison, danger	Relational threat ontology and separate chemical adversity from discernment.	Core Ontology; Chemical Domain Boundaries
Reliable map from unreliable information	Map types, calibration, active sampling, remapping, held-out tests.	Experiment Families; Measures
Different sensory apparatuses	Anti-anthropocentric sensorium profile and entity-specific apparatus validation.	Laboratory Architecture
Threats not universal or constant	No-threat baselines, context indexing, protected variables, time horizon.	Purpose and Scope; Core Ontology
Threats may be undetectable, unknowable, or unaddressable	Detectability audit, unknown state, open-world trials, actionability dissociation.	Foundational Propositions; Experiment Families
Laboratory cannot test all threats	Bounded claim language, scope register, uncertainty disclosure, no universal safety certification.	Purpose and Scope; Ethics
Need alternative analysis	Mandatory competing lenses and discriminating predictions.	Alternative Analyses
Need version record	Document identifier, version log, roles, traceability, revision rules.	Document Control

Evidence Notes

These sources inform the framework; they do not convert the concept into an approved protocol.

[1] Brebner, J. S., et al. (2024). "Through an animal's eye: the implications of diverse sensory systems in scientific experimentation." Proceedings of the Royal Society B, 291:20240022. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11253838/> — Supports species-specific sensory design, anti-anthropocentric controls, ecological context, and reporting of hidden cues.

[2] Kayser, C., & Shams, L. (2015). "Multisensory Causal Inference in the Brain." PLOS Biology, 13:e1002075. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4339834/> — Supports the distinction between cue integration and inference about common versus separate causes.

[3] "Temporal Binding in Multisensory and Motor-Sensory Contexts: Toward a Unified Model" (2021). Frontiers in Human Neuroscience. <https://www.frontiersin.org/articles/10.3389/fnhum.2021.629437/full> — Supports probabilistic temporal binding and precision-weighted cue combination.

[4] U.S. EPA. Ecological Risk Assessment. <https://www.epa.gov/risk/ecological-risk-assessment> — Distinguishes planning, problem formulation, exposure/effects analysis, and risk characterization under uncertainty.

[5] U.S. EPA. Conducting an Ecological Risk Assessment. <https://www.epa.gov/risk/conducting-ecological-risk-assessment> — Supports explicit entities, endpoints, conceptual models, exposure paths, confidence, uncertainties, and alternative interpretations.

[6] OECD. Guidance Document for the Use of Adverse Outcome Pathways in Developing Integrated Approaches to Testing and Assessment (IATA). https://www.oecd.org/en/publications/guidance-document-for-the-use-of-adverse-outcome-pathways-in-developing-integrated-approaches-to-testing-and-assessment-iata_44bb06c1-en.html — Supports integrating alternative methods and mechanistic knowledge for chemical assessment.

[7] NC3Rs. “What are the 3Rs?” <https://nc3rs.org.uk/> — Replacement, Reduction, and Refinement guide the ethical use of animals in research and testing.

[8] ARRIVE Guidelines 2.0, Randomisation. <https://arriveguidelines.org/arrive-guidelines/randomisation/4a/explanation> — Supports randomization, blocking, confound control, and transparent allocation reporting.

[9] Percie du Sert, N., et al. (2020). ARRIVE Guidelines 2.0 and explanation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7360025/> — Supports transparent reporting, blinding disclosure, exclusions, and protocol planning.

[10] Boulton, T. E., et al. (2019). “Learning and the Unknown: Surveying Steps toward Open World Recognition.” AAAI. <https://aaai.org/papers/09801-learning-and-the-unknown-surveying-steps-toward-open-world-recognition/> — Supports distinguishing unknown inputs from mere low confidence and notes unresolved limitations.

[11] Pezzulo, G., et al. (2015). “Active Inference, homeostatic regulation and adaptive behavioural control.” <https://pmc.ncbi.nlm.nih.gov/articles/PMC4779150/> — Provides one candidate account connecting prediction error, action, and homeostatic control.

[12] U.S. OSHA. Hazard Communication Standard and Appendix A. <https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.1200> and <https://www.osha.gov/hazcom/appendix-a> — Supports separating hazard classes and distinguishes irreversible corrosion from reversible irritation.

[13] “The History and Philosophy of Ecological Psychology” (2018). <https://pmc.ncbi.nlm.nih.gov/articles/PMC6280920/> — Supports affordances as organism-relative possibilities for action.

Approval Record

Role	Name	Decision	Date	Signature / record
Originator / Research Architect		Pending		
Scientific reviewer		Pending		
Entity-domain reviewer		Pending		
Ethics / welfare reviewer		Pending		
Chemical safety / biosafety reviewer		Pending		
Methods / statistics reviewer		Pending		
Version approver		Pending		

Approval of this foundational document does not approve any experiment. Each implementation requires its own protocol, risk assessment, ethics review, and operational authorization.